1. Clinical Question & Core Hypothesis

Clinical Question

Organized intratumoral immune structures (TLS) or heavy lymphoid infiltration should limit progression, yet some immune-infiltrated tumors still progress—why? 2) Tier 1 paths used: (a) PSMA is expressed in cancer stem–like cells (path 3), providing a candidate tumor-intrinsic, therapy-resistant cell population; (b) PI3K signaling is functionally linked to PD-L1–mediated programs (path 6), suggesting a PD-L1↔PI3K axis that could enable immune escape. I built the core hypothesis on the intersection of stemness (PSMA+) and immune-checkpoint signaling (PD-L1). 3) Tier 2 supporting evidence: TMPRSS2/ERG cooperativity with PI3K (path 5) supports that oncogenic programs can amplify PI3K-driven immune-evasive signaling; BCG/PD-L1 links (paths 7,10,15) illustrate that PD-L1 is a dynamic, therapy-modulated node, reinforcing PD-L1's role in shaping immune outcomes. 4) Why PFI chosen as endpoint: progression-free interval is most sensitive to early relapse/progression driven by therapy-resistant stem-like populations and thus directly tests whether PSMA+/PD-L1–mediated mechanisms allow tumor progression despite organized immune infiltrates. OS is more confounded by later-line treatments; therapy-response proxies in TCGA-BRCA are limited. 5) Deliberate exclusions: I excluded TMPRSS2/ERG and PIK3CA from the core anchors to keep the mechanistic claim compact (max 2 anchors); androgen-deprivation and BCG are contextually relevant but not central to breast cohorts in TCGA-BRCA, so they are placed outside the core hypothesis to avoid overfitting the claim to cross-tissue inferences?

Working Hypothesis

In TLS-rich breast tumors, a subpopulation of PSMA (FOLH1)-expressing cancer stem–like cells activates a PD-L1–dependent PI3K-AKT signaling axis, promoting immune evasion and continued tumor progression despite organized TLS. Disrupting PSMA, PD-L1, or PI3K signaling will reduce this immune-evading program and extend progression-free interval (PFI).

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PSMA, PD-L1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PSMA: PSMA is expressed in cancer stem–like cells (path 3), providing a candidate tumor-intrinsic, therapy-resistant cell population; (b) PI3K signaling is functionally linked to PD-L1–mediated programs (path 6), suggesting a PD-L1↔PI3K axis that could enable immune escape.

PD-L1: PD-L1 links (paths 7,10,15) illustrate that PD-L1 is a dynamic, therapy-modulated node, reinforcing PD-L1's role in shaping immune outcomes.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	TMPRSS2/ERG	Supporting modifier from knowledge graph
Secondary	PIK3CA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: perform correlative analyses of PSMA/FOLH1, PD-L1/CD274, TLS gene signatures, and PFI;

Prediction 2: engineer PSMA overexpression/knockdown and PD-L1 modulation in breast cancer cell lines and organoids, assessing PI3K-AKT signaling and immune-killing in co-cultures;

Prediction 3: test TLS formation and tumor control in humanized mouse models with PSMA/PD-L1/PI3K perturbations and PD-1/PD-L1 blockade;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TMPRSS2/ERG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PIK3CA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.